Spring 2026

An Analysis of the Clinical Factors Leading to an Increase in Traumatic Injury

Caitlin Roake, Laura Sinden, Calder Lake, Tyler Huyette, Stephanie Ehlert

BMI 6016

Background & Motivation

We wanted to look at a database to explore the factors contributing to increased trauma burden in the Emergency Department. We predict that older adults with trauma will have increased severity of trauma

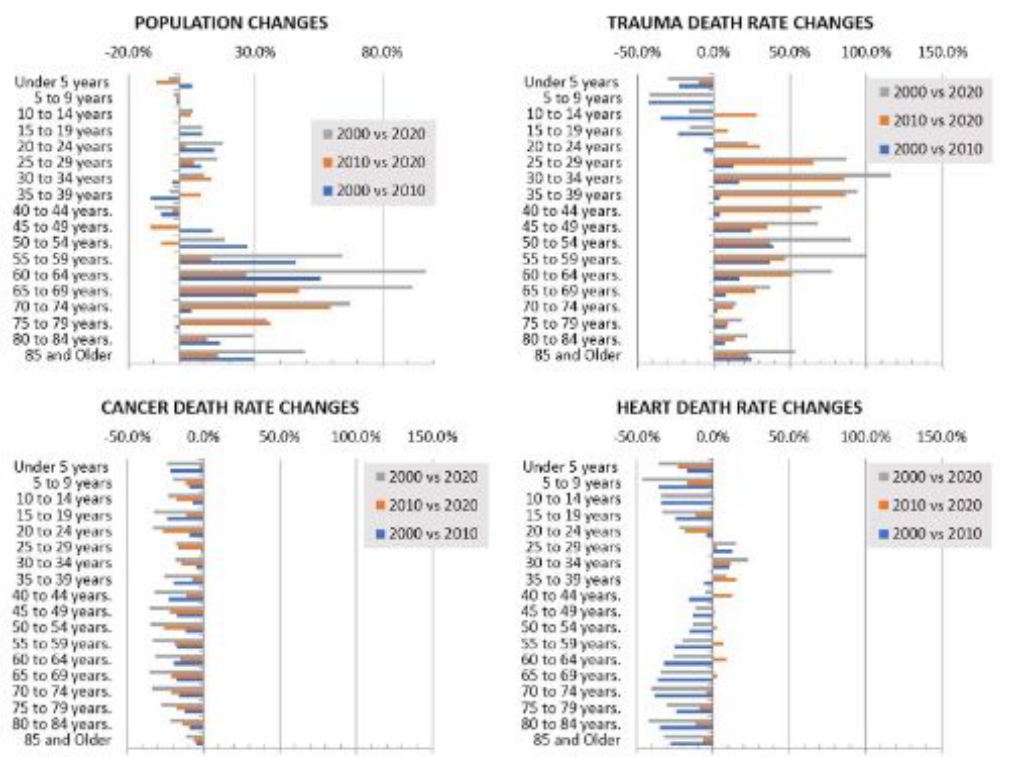


FIGURE 4. Changes in population and death rates by trauma, cancer, and heart disease by age groups and by years 2000, 2010, and 2020.

Project Objectives

Comparison groups:
Adults 18-65, Adults 65+

Primary Outcome:

1. Comparison of the trauma injury mechanism (fall, firearm, assault) from ICD-10 codes between young adults and older adults

Secondary outcome:

2. Comparison of trauma severity between young adults and older adults (as measured by shock index)

Ethics

This retrospective study is exempt from informed consent.

The data reviewed is a secondary analysis of existing data, does not involve intervention or interaction with human subjects, and is de-identified per the de-identification standard defined in Section §164.514(a) of the HIPAA Privacy Rule.

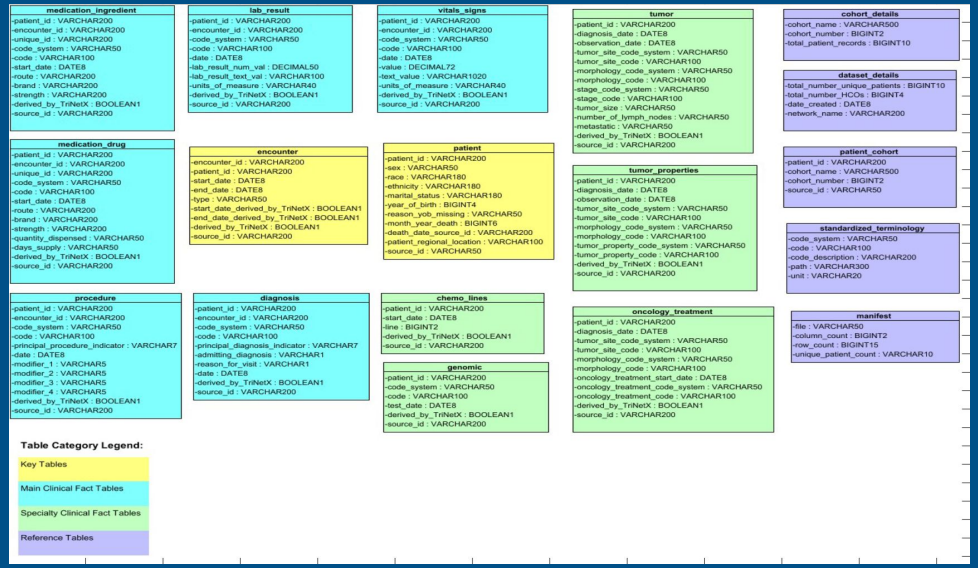
Data

The data is derived from TriNETX with the following parameters:

1. Emergency Department visit
2. Adults (age > 18)
3. Severe mechanism of injury (defined as trauma to the head, chest, or abdomen)

This returned about 7.6 million patient records from 101 healthcare organizations for which we have the patient table and the diagnosis table.

We selected diagnoses with injury-based ICD-10 codes("V", "W", "X", "Y"). We did not use ICD-9 codes in this analysis. We filtered those diagnoses that were associated with an emergency room visit. This resulted in a dataset of about 121,000 unique patients.



ICD madness

In our initial analysis we initially miscoded ICD9 'V' codes as traffic accidents

As a result most of our motor vehicle accident data was actually related to pregnancy

A SQL based ICD10 search through the TriNetX diagnoses file did not reveal any 'V' (motor vehicle accident) codes

```
 code_letter       n
0           W  331578
1           X  105401
2           Y  355182
```

Mechanism of injury	ICD-9 codes	ICD-10 codes
Included in study		
Falls	E880-E888	W00-W19
Injuries to pedal cyclists	E810-E819 (with 4th digit .6) E826-E829 (with 4th digit .1)	V10-V19
Injuries to pedestrians	E810-E819 (with 4th digit .7) E826-E829 (with 4th digit .0)	V01-V09
Other transport injuries	E800-E807, E810-E819 (with 4th digit .0-.5, .8-.9) E820-E825, E826-E829 (with 4th digit .2-.4, .8-.9) E830-E848	V20-V99
Injuries from fire, smoke, and hot objects or fluids	E890-E899 E924 (with 4th digit .0, .8)	X00-X19
Poisoning or contact with corrosive substances	E850-E858 E860-E869 E924 (with 4th digit .1)	X40-X49
All other unintentional injury mechanisms	E900-E923 E924 (with 4th digit .9) E925-E928 E980-E989	W20-W45 W49-W60 W64-W94 W99, X20-X39 X50-X59 Y10-Y34 Y90-Y98

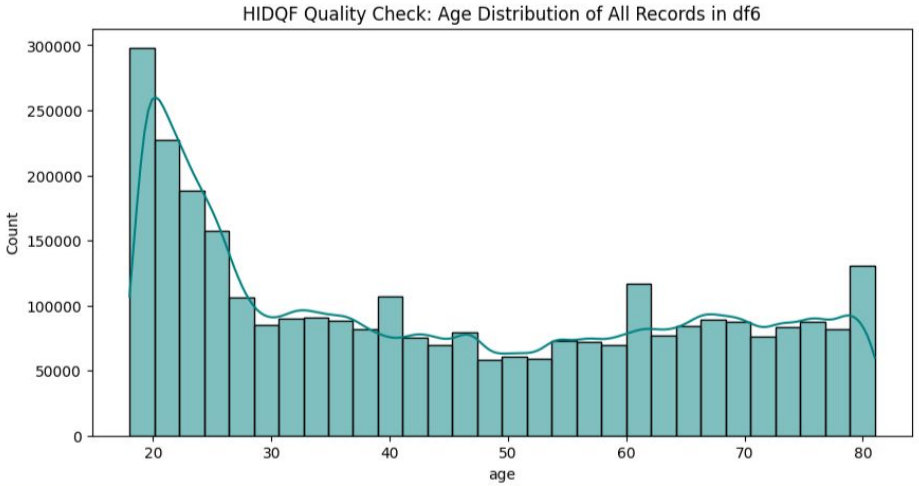
Data Descriptive

1. On first analysis of the data there were a lot of duplicate records, so we cleaned the data:
   - Total records found: **1001643**
     Total unique patient-trauma instances: **88109**

	patient_id	code	age
0	vB2wD	W20.8XXA	31
5	vh2wD	W19.XXXA	73
13	vx2wD	W19.XXXA	35
31	vh3wD	W19.XXXA	57
45	vh4wD	W18.09XA	33

   - The cleaned data looked good after the quality check

```
--- Completeness Check ---
Missing Diagnosis Codes: 0
Missing Patient IDs: 0
Missing Birth Years: 0

--- Plausibility Check ---
Records with impossible ages (<0 or >115): 0
```



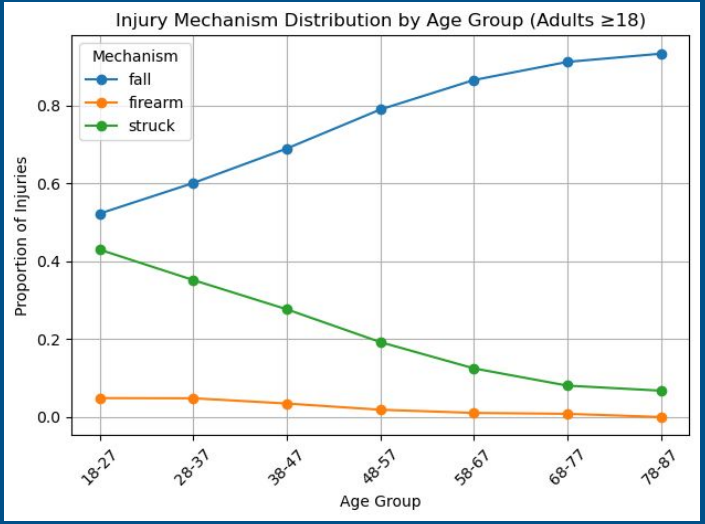
Data Descriptive

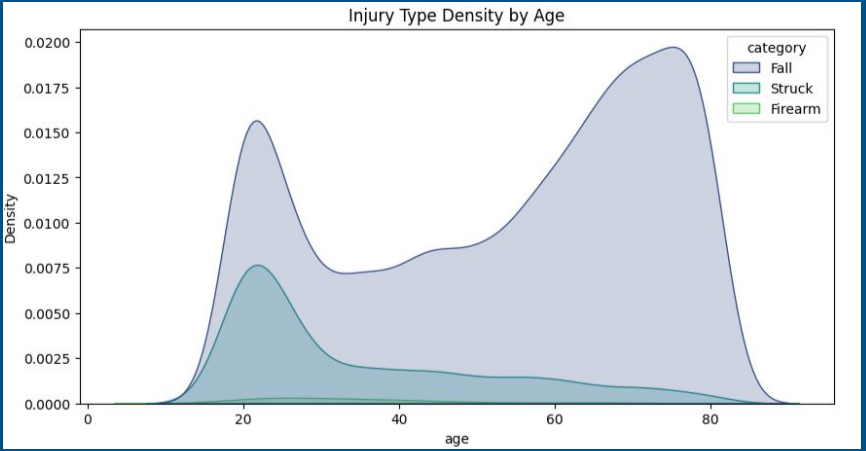
2. Number of unique patients with trauma diagnoses:.

```
Breakdown by Trauma Type:
Fall        68128
Struck      18765
Firearm      1216
```

Average Age by Trauma Type:

```
Fall        51.272326
Firearm     36.226151
Struck      38.587317
```

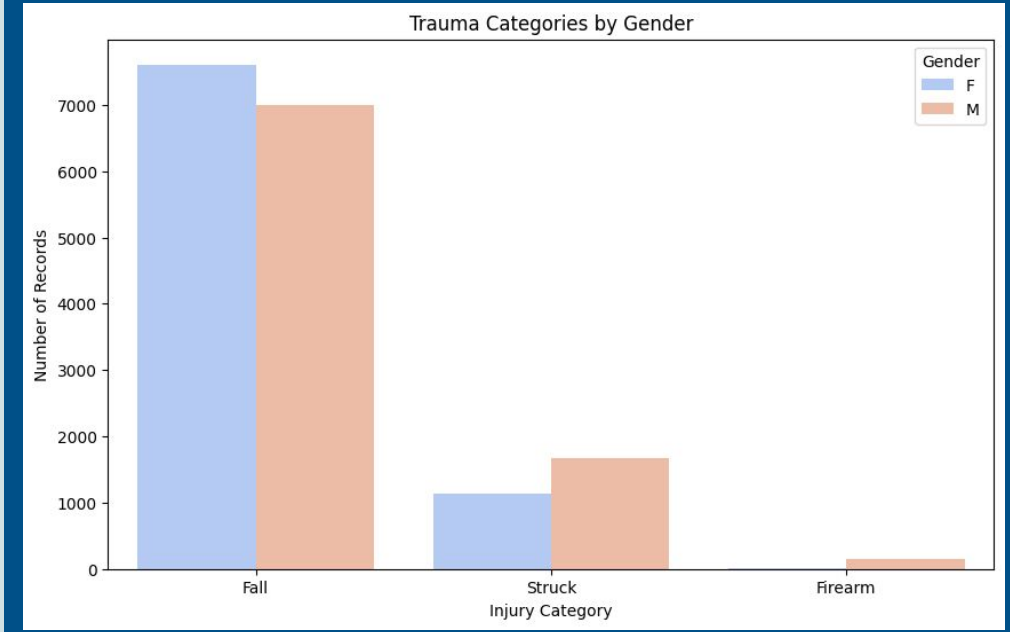




Data Descriptive

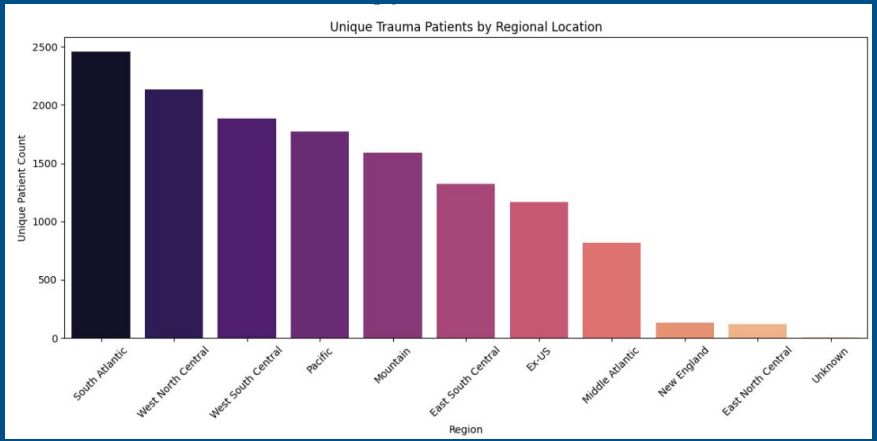
3. Sex of the patients with a trauma diagnosis:

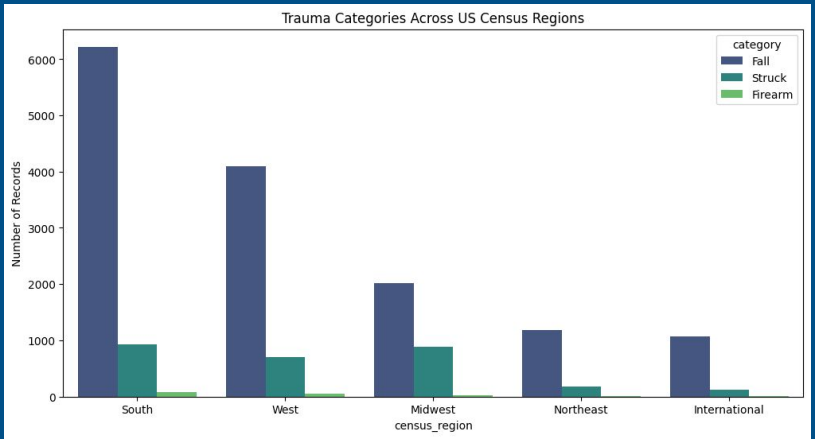
```
--- Percentage of Injury Types Within Each
Gender —

category   Fall  Firearm  Struck
sex
F         86.87     0.21   12.93
M         79.36     1.66   18.99
```



Data Descriptive

Regional Distribution Summary- We attempted to get cases by state but were unable to because TriNetX typically organizes U.S. data by census divisions to provide geographic context while maintaining privacy at scale.





Data Processing

1. We will assign trauma ICD-10 codes to three mechanism categories: Falls, Struck, and Firearm injuries. LOINC codes were used to calculate shock index from vital signs

- Falls
  W00 - W19
  X80, Y30
  175 Total Codes

- Struck
  W20 - W22
  W50 - W52
  X79, Y29
  87 Total Codes

- Firearms
  W32 - W34
  X72 & X74
  Y22 - Y24
  36 Total Codes

- LOINC codes
  Systolic BP: 8480-6
  Heart Rate: 8867-4

Data Processing

2. We will assign patient demographics (patient id, age) to the mechanism & injury type diagnoses either using SQL or pandas dataframes

```
[5]: df_final.columns

[5]: Index(['patient_id', 'encounter_id', 'code_system', 'code', 'patient_id_1',
       'sex', 'race', 'ethnicity', 'marital_status', 'year_of_birth',
       'reason_yob_missing', 'month_year_death', 'death_date_source_id',
       'patient_regional_location', 'source_id', 'patient_id_2',
       'encounter_id_1', 'code_system_1', 'code_1', 'date', 'value',
       'text_value', 'units_of_measure', 'derived_by_TriNetX', 'source_id_1',
       'code_clean', 'fall', 'firearm', 'struck', 'injury_type',
       'year_of_visit', 'estimated_age', 'age_group', 'sbp', 'hr',
       'shock_index'],
      dtype='str')
```

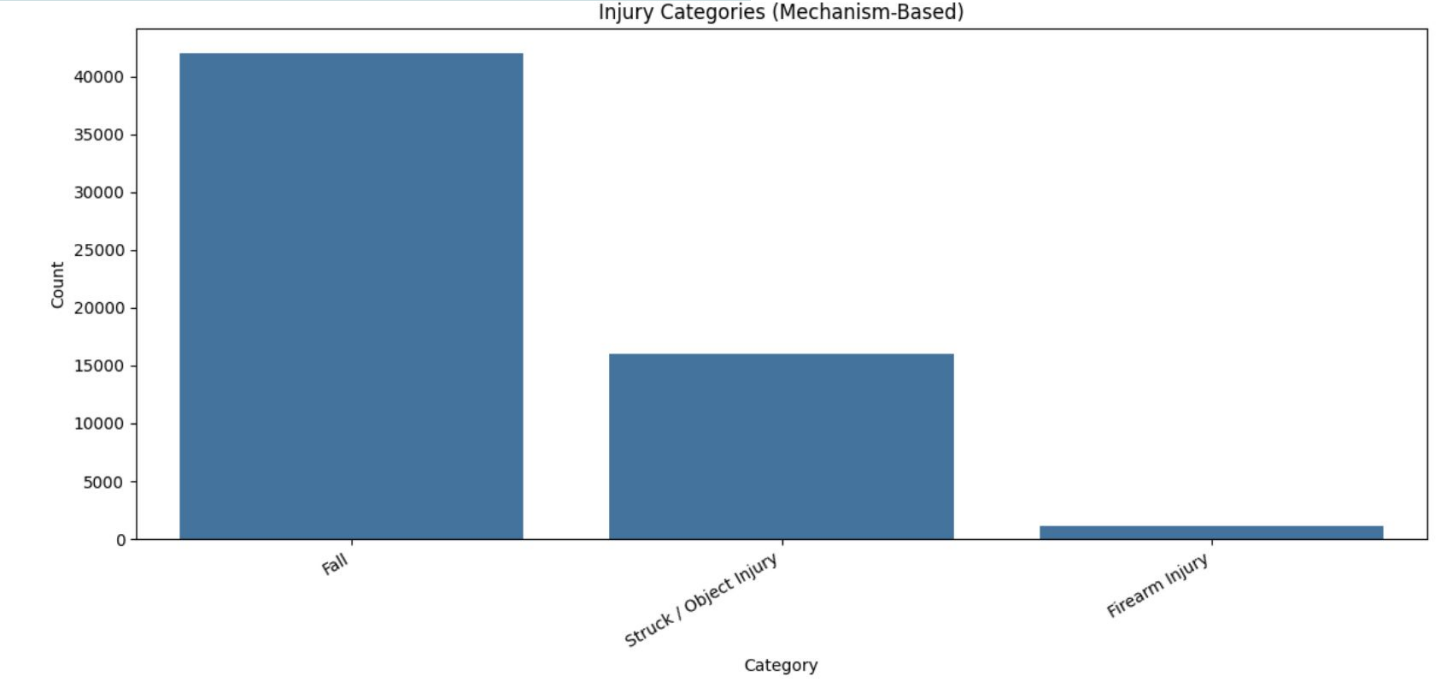
Data Visualization

1. Trauma counts by injury category
2. Heatmap of regional distribution of trauma injuries
3. Trauma type distributions by age
4. Injury categories & proportions by region
5. Trauma probability by age & region
6. Regression charts of probability for injury type with respect to age and location

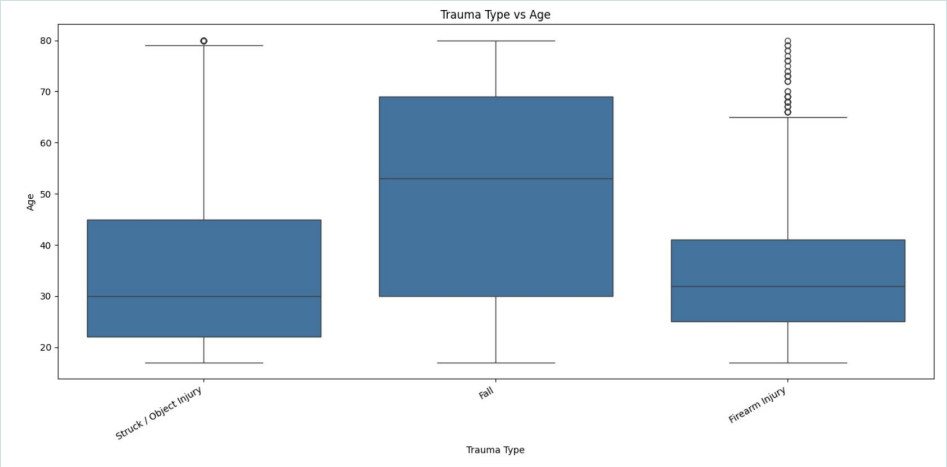
Trauma Injury Counts by Category

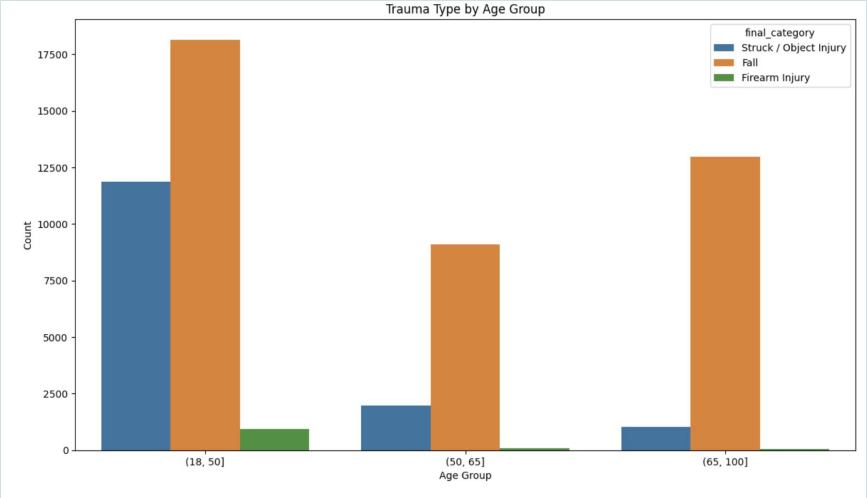
```
trinet_df['mechanism_category'].value_counts()
```

```
mechanism_category
Fall                        42023
Struck by/Against Object    16015
Firearm Injury               1132
Name: count, dtype: int64
```

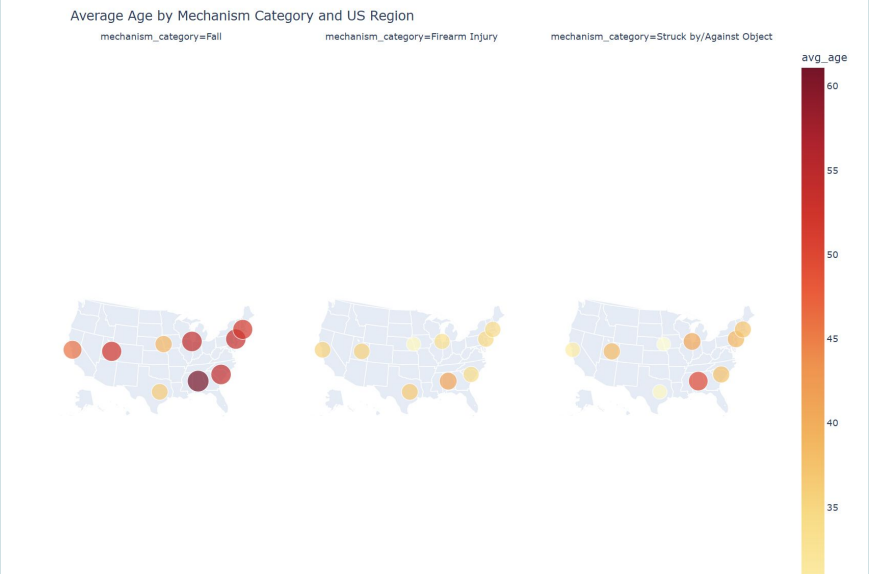


Trauma Type Distribution by Age

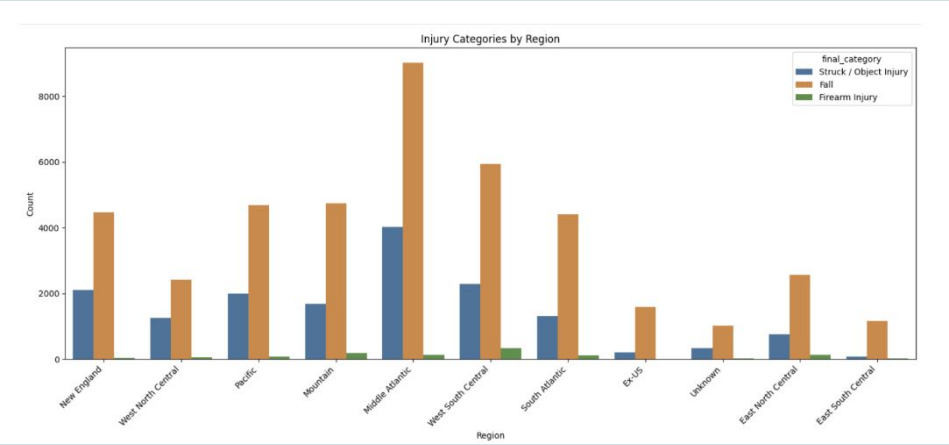


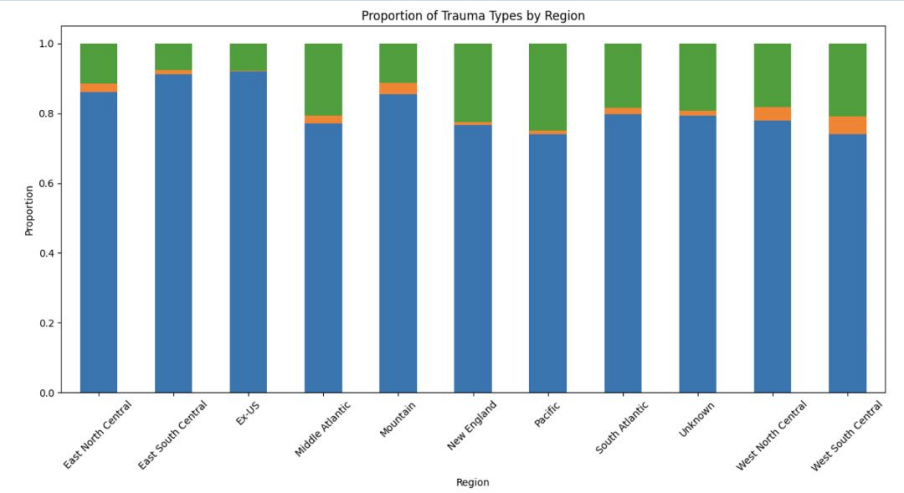


Average Age By Trauma Type & Region



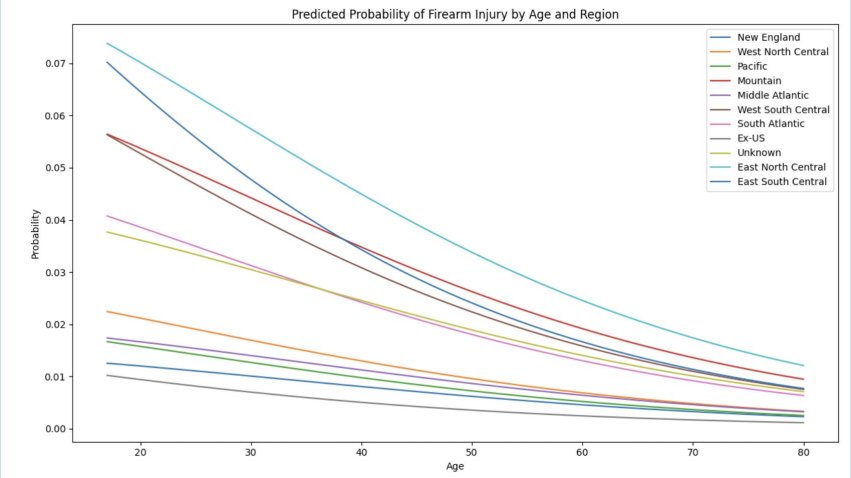
Trauma Injury Categories & Proportions by Region

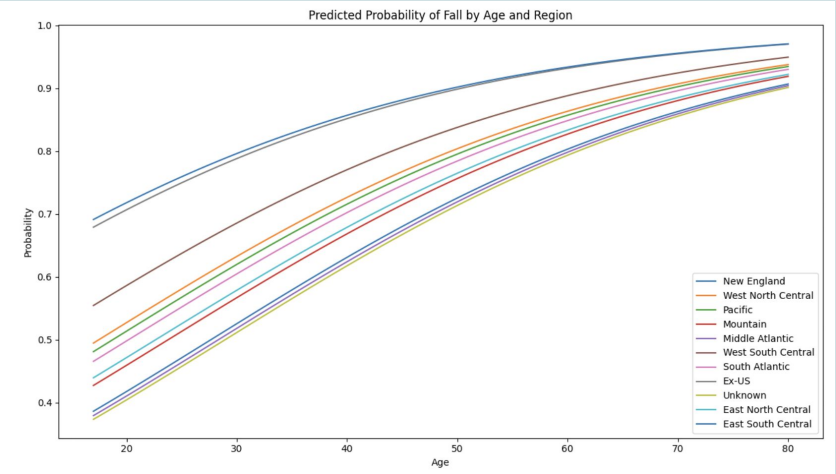


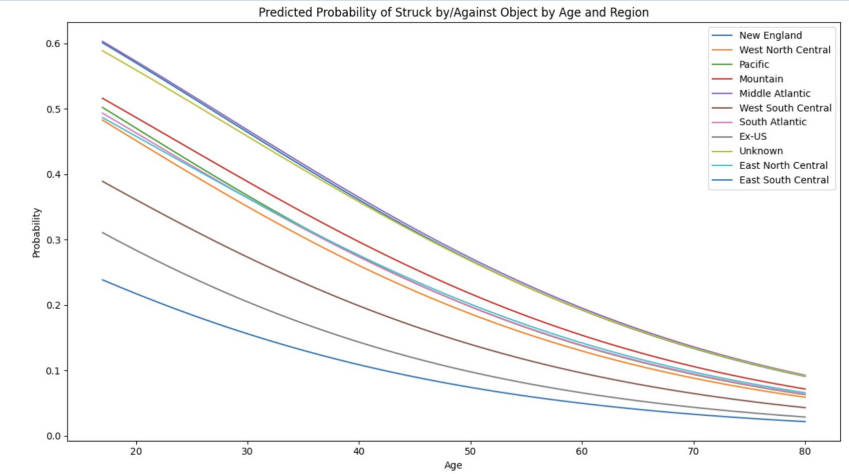


BMI 6016

Trauma Type Probability by Age & Region







Discussion

- Data can be messy! We had issues:
  - Slow to get the data from TriNETX
  - Difficult getting access to Narwhal (3 of us gave up)
  - Finding the right ICD-10 codes across multiple spreadsheets
  - Missing data for Motor Vehicle Accidents (we just used falls, firearms and assault)

Discussion

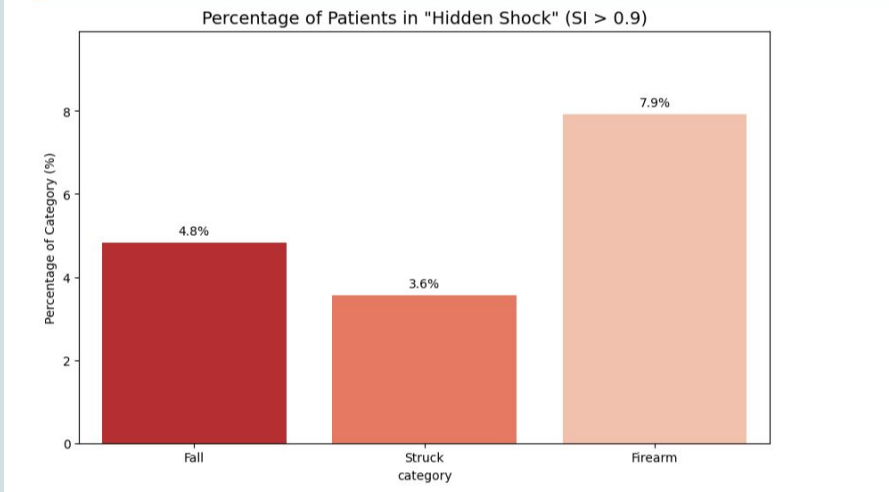
The formula for Shock Index (SI) is:
Heart Rate(bpm)
Systolic Blood Pressure (mmHg)

Normal Range 0.5 - 0.7
Greater than 0.9 is defined as shock
The Shock Index is used to determine how sick a patient is when arriving in the ED

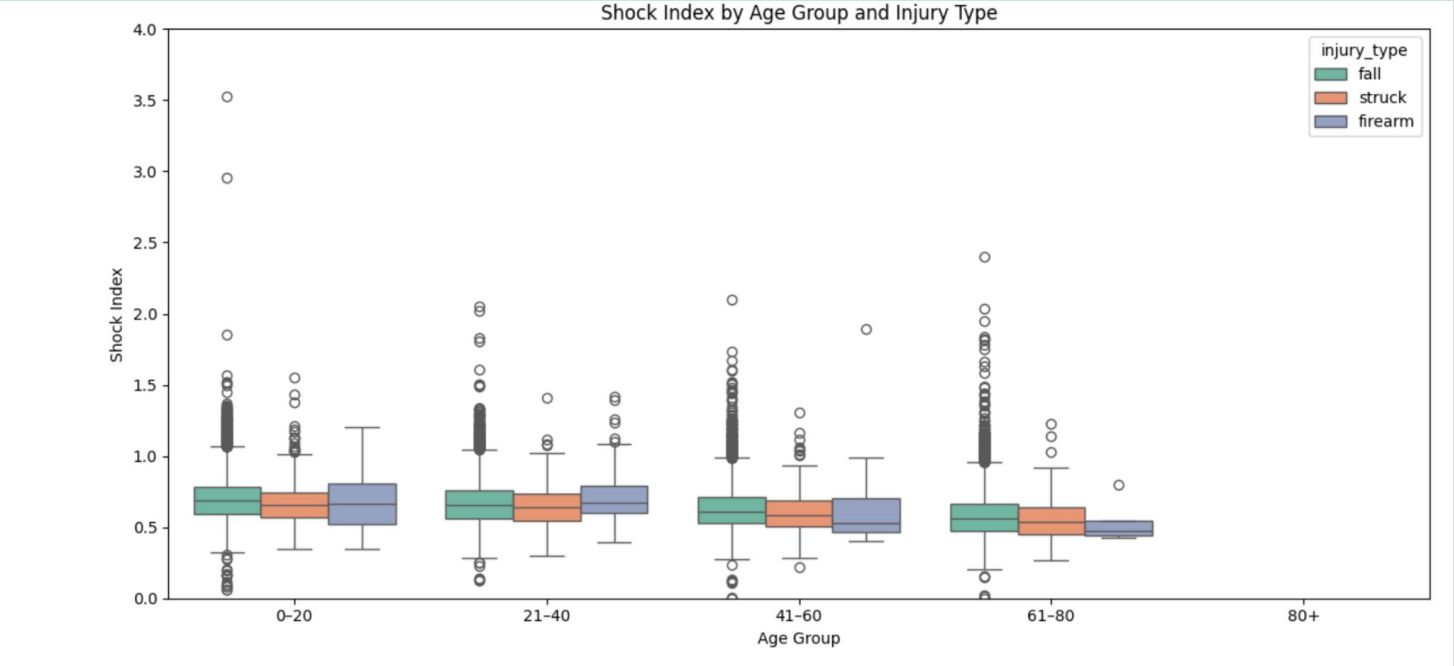
- Patients in Hidden Shock *(>0.9) ---

Category	Total Patients	Patients in Shock (>0.9)	Percentage (%)
Firearm	164	13	7.926829
Fall	14599	705	4.829098
Struck	2805	100	3.565062

*Note-We called it "Hidden Shock" because most emergency room encounters did not result in a value >0.9

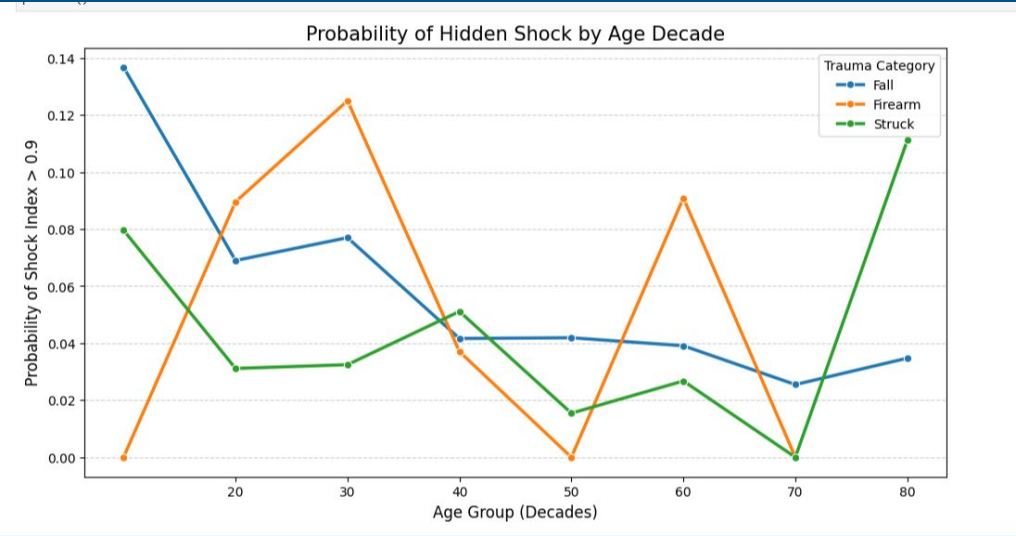


Shock Index by Age Group & Injury Type



Discussion

- It appears age changes how the body reacts to trauma.
- For most injuries, like falls and firearms, younger patients are much more likely to have a high shock index.
- As patients get older, that 'warning light' seems to fade out, even if the injury is serious.
- This is a big deal because it means older patients might actually be in trouble even if their vitals look okay on paper.
- The one big exception is in the 80s age group—when they get 'struck' by something, their risk of shock shoots way up compared to younger people.



References

1. Rhee P, Holcomb JB, Zangbar B. Evolving Epidemiology of Increasing Trauma Deaths in the United States (2000-2020). Ann Surg. 2025 Jun 1;281(6):976-981

2. Tomas C, Kallies K, Cronn S, Kostelac C, deRoon-Cassini T, Cassidy L. Mechanisms of traumatic injury by demographic characteristics: an 8-year review of temporal trends from the National Trauma Data Bank. Inj Prev. 2023 Aug;29(4):347-354. doi: 10.1136/ip-2022-044817